EEC 4414 / UNDERGRADUATE

Deep Learning

How neural networks learn, and how their architectures handle images, sequences, graphs, and generative tasks.

COURSE CONTENT

Neural-network foundations	Tensors, automatic differentiation, linear regression, softmax classification, and multilayer perceptrons.
Training and generalization	Backpropagation, initialization, gradient-based optimization, weight decay, dropout, and overfitting.
Convolutional networks	Convolutions, pooling, LeNet, residual networks, and batch normalization.
Sequences and attention	RNNs, LSTMs, GRUs, encoder-decoder models, attention, and Transformers.
Generative models	Variational autoencoders, diffusion, text-to-image generation, guidance, and sample-quality evaluation.
Learning with fewer labels	Data augmentation, transfer learning, contrastive and semi-supervised learning, and meta-learning.
Graph neural networks	Message passing, graph convolutions, GraphSAGE, and graph attention.

LEARNING FOCUS

Relate architectures and loss functions to training behavior, data structure, and task requirements.

Mathematical explanations, architecture diagrams, and Python/PyTorch examples.